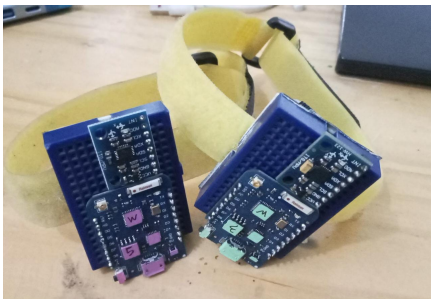
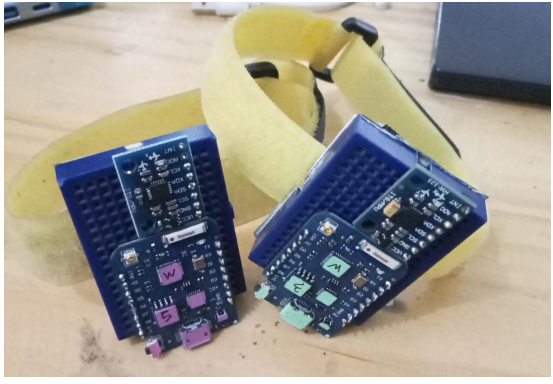
CLASSROOM DEMONSTRATION / WEARABLE IMU TRACKING

Poor Man’s IMU-Based Motion Tracking

3D arm tracking with two ESP8266 boards, MPU6050 sensors and Madgwick fusion



What we built

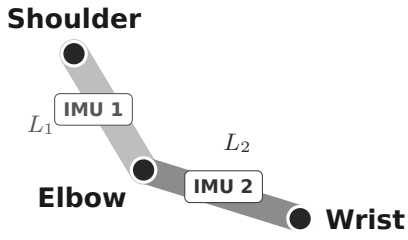


HARDWARE PER NODE

ESP8266 + MPU6050

- Two independently measured arm segments
- Gyroscope + accelerometer, 6 axes each
- One USB serial connection per node
- Laptop estimates 3D joint positions

Two IMUs, two rigid segments

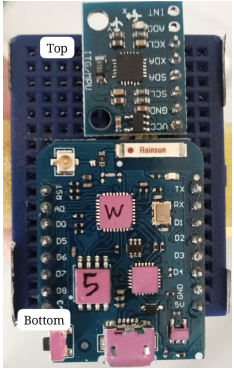


THE TRACKING MODEL

One orientation estimate per bone

- Upper arm and forearm are treated as **separate rigid bodies**.
- Two quaternions orient two known bone-direction vectors.
- The shoulder is a fixed visualization anchor.
- The wrist is a **3D position**, not a full hand pose.

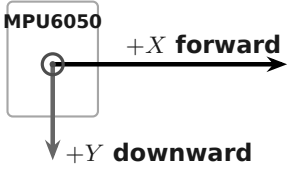
Left-arm sensor orientation



Actual node: IMU at the top; USB connector at the bottom.

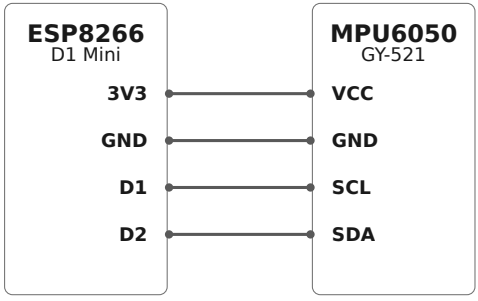
LEFT ARM: STRAIGHT-DOWN REFERENCE POSE

Sensor axes after mounting



- $+Z$ points outward to the left.
- Both sensors must have the same orientation.

Wiring: ESP8266 to MPU6050



I^2C: SDA = GPIO4, SCL = GPIO5

FOUR CONNECTIONS PER SENSOR

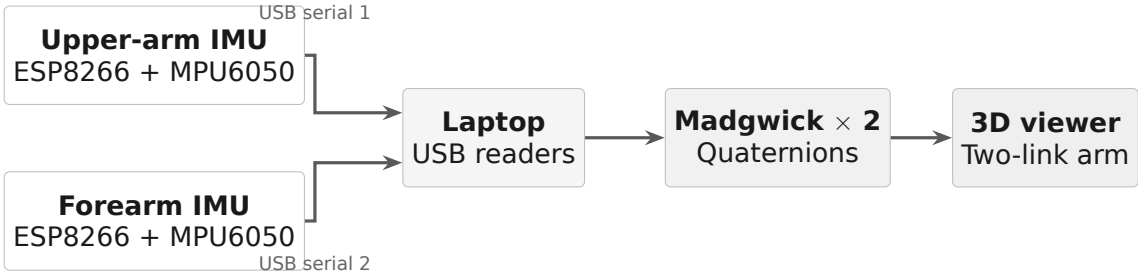
ESP8266	MPU6050
3V3	VCC
GND	GND
D1 (GPIO5)	SCL
D2 (GPIO4)	SDA

Power: 3.3 V (not 5 V).

Address: `0x68` (default).

Repeat for both nodes. Each ESP8266 uses its own USB serial connection to the laptop.

Two USB streams, one laptop



Demo setup: both ESP8266 boards run identical firmware and stream raw IMU data at a nominal 100 Hz target. The laptop performs fusion and graphics; no Wi-Fi link is required.

Why Madgwick fusion?

GYROSCOPE

Fast motion

Integrate angular rate to estimate orientation.

Bias causes drift.

ACCELEROMETER

Gravity reference

Gravity provides a tilt reference.

Motion distorts tilt.

MADGWICK

Quaternion fusion

Combines gyro prediction with gravity correction.

3D tilt, relative heading.

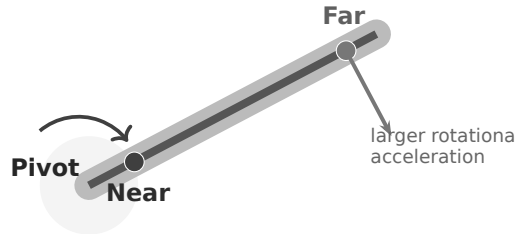
Important: MPU6050 has no magnetometer. Absolute heading (yaw about gravity) is unobservable, so the independently estimated headings can drift.

Near-joint IMU placement

RIGID-BODY ANGULAR VELOCITY

$$\boldsymbol{\omega}_A = \boldsymbol{\omega}_B$$

For any two points on the same rigid segment, the angular velocity vector is the same when expressed in a common frame.



BUT LINEAR ACCELERATION DEPENDS ON POSITION

$$\mathbf{a}_B - \mathbf{a}_A = \boldsymbol{\alpha} \times \mathbf{r}_{AB} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{AB})$$

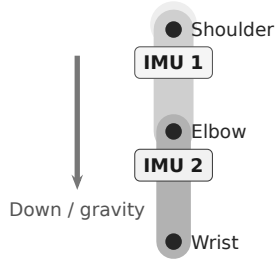
At $\omega = 5\,\text{rad/s}$, $a_c = \omega^2 r$:

$r = 2\,\text{cm} \Rightarrow a_c = 0.5\,\text{m/s}^2$

$r = 20\,\text{cm} \Rightarrow a_c = 5.0\,\text{m/s}^2$

Calibration and initial reference

1. Strap each module firmly to its own segment.
2. Hold both stationary at startup to estimate gyro bias.
3. Stand upright with the left arm straight **down**.
4. Use this pose to set the relative zero orientation; yaw is not observable.
5. Press **C** in the viewer, then move the arm slowly.



3D forward kinematics

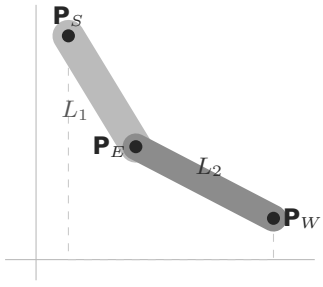
STEP 1: ROTATE EACH BONE DIRECTION

$$\mathbf{d}_i = R(q_i)\,\mathbf{b}_i, \qquad i \in \{1, 2\}$$

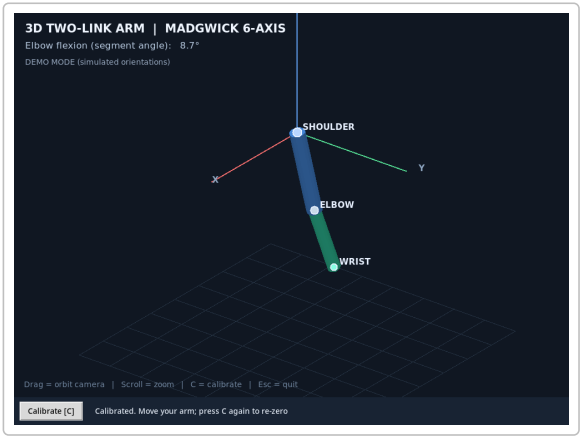
q_i: aligned sensor orientation; $\mathbf{b}_i$: calibrated sensor-to-bone direction.

STEP 2: FORWARD KINEMATICS

$$\mathbf{P}_S = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^\mathsf{T}$$
$$\mathbf{P}_E = \mathbf{P}_S + L_1 \mathbf{d}_1$$
$$\mathbf{P}_W = \mathbf{P}_E + L_2 \mathbf{d}_2$$



3D visualization with cylinders



VIEWER FEATURES

- 3D orbit and zoom
- Distinct upper and lower segments
- Elbow flexion indicator
- Per-sensor sample status
- **C** to re-zero the pose

Three classroom experiments

EXPERIMENT 1

Slow vs. fast motion

Compare the sensor readings at rest, during slow movement, then during fast movement.

Lesson: movement distorts gravity-based tilt.

EXPERIMENT 2

Sensor placement

Move an IMU toward or away from a joint while keeping it on the same rigid segment.

Lesson: same gyro motion, different acceleration.

EXPERIMENT 3

Drift and re-zeroing

Wait, observe the pose, return to zero, and press C to recalibrate.

Lesson: yaw drifts without heading reference.

Limitations and next steps

CURRENT LIMITATIONS

- **Yaw drift:** no magnetometer or external heading reference
- **Strap motion:** soft tissue is not perfectly rigid
- **Dynamic acceleration:** tilt correction can be disturbed
- **Timing:** serial streams are not perfectly synchronized
- **Scope:** two-link arm only, no hand pose

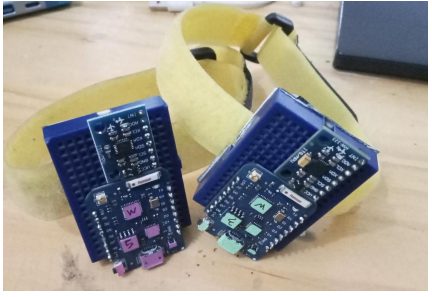
POTENTIAL IMPROVEMENTS

- Add packet timestamps and robust stream recovery
- Improve IMU-to-bone alignment and per-user lengths
- Compare raw gyro integration with fusion
- Add anatomical elbow constraints if needed
- Replace USB with wireless links in a future wearable

YOUR FINAL PROJECT

Sensors → Fusion → Kinematics → CoppeliaSim

- **Task 1: Virtual Arm**
  Animate a virtual human arm in CoppeliaSim using real-time IMU data.
- **Task 2: Rehabilitation**
  Develop an interactive rehabilitation exercise, such as target reaching.
- **or Task 2: Motion Assessment**
  Measure joint range of motion, movement duration, and exercise repetitions.



You can use more IMUs!